

Detector Design Spec 2 — 4H-SiC Microdosimeter (sensitive-volume geometry)

For: foundry hand-off (FBK Schottky / ST JBS / other) From: Petringa group, INFN-LNS — 4H-SiC TCAD study
Date: 2026-06-16 Status: GEOMETRY-ONLY physics specification (NOT a mask set; junction type NOT modeled)

Scope and honesty boundary. This document specifies the sensitive-volume geometry and electrode layout of a 4H-SiC microdosimeter, TCAD-optimized for charge-collection uniformity. It does NOT specify the junction type or its electrical behavior. Our simulator has no Schottky and no JBS model — so the barrier height, built-in potential, leakage, and C-V of a metal/JBS front contact are explicitly left to the foundry (FBK Schottky, ST JBS). The geometry is portable across junction types; the depletion-bias numbers below assume a p-n-like junction electrostatics and must be re-derived by the fab for a Schottky/JBS contact (see §4).

1. Device concept

Array of small sensitive volumes (SVs) in a 4H-SiC epitaxial layer, each read out through a front electrode. The microdosimetric geometry (SV dimensions, depth, electrode layout, CCE uniformity across the SV) is what we specify. The junction that forms the collecting contact — Schottky (FBK) or junction-barrier-Schottky (ST) — is the foundry's choice and is dropped into this geometry.

2. Sensitive-volume geometry spec (TCAD-optimized, p-n electrostatics assumed)

Parameter	Recommended value	Basis / note
SV depth (epi thickness)	10 μm (baseline)	calibrated/validated config
SV half-width	50 μm ($\rightarrow$ 100 μm SV)	the width we actually validated for uniformity
Graded epi donor profile	N_D_junction 2.93e15, N_D_bulk 8.82e13, L 0.99 μm	calibrated (Phase 26)
N_D(y)		
Operating reverse bias	-15 to -20 V	margin above V_fd, well below the -50 V solver envelope
Full-depletion voltage V_fd (10 μm)	$\approx$ 10.5 V	calibrated profile (assumes p-n electrostatics)
CCE-uniformity acceptance	edge & center CCE $\geq$ 0.95, fully depleted	Mj-3 validity gate

Alternative SV depths (verified V_fd, p-n electrostatics):

Epi thickness	V_fd	Recommended operating bias	Note
5 μm	$\sim$ 4.1 V	$\geq$ -10 V	lowest-bias option
10 μm	$\sim$ 10.5 V	-15 to -20 V	baseline (most validated)
20 μm	$\sim$ 35 V	$\geq$ -40 V	convergent but less margin to -50 V envelope

Validity gate (audit Mj-3): only configurations that are fully depleted ($|V_{\text{bias}}| \geq V_{\text{fd}}$) and above the CCE floor (0.95 at both center and edge) are quoted. A partially-depleted SV can show a deceptively uniform edge/center ratio at low (both-poor) CCE — those are excluded.

3. What the geometry deliverable gives the fab

- SV depth, half-width, and the laterally-uniform graded doping target for which CCE is uniform across the SV at the operating bias.
 - Full-depletion voltage and recommended operating bias for a p-n-type junction.
 - Confirmation that $CCE \rightarrow 1$ (> 0.96) across the SV when fully depleted (DD model, validated vs 1D twin).
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4. Junction type — NOT modeled (fab fills in), and why the bias numbers shift

The collecting junction (FBK Schottky / ST JBS) is not in our simulator. Specifically NOT provided and to be supplied/measured by the fab:

- Schottky/JBS barrier height ϕ_B , built-in potential, image-force lowering.
- Leakage of the metal/JBS junction (Schottky leakage $\gg$ p-n; thermionic + TFE).
- C-V and the bias needed to fully deplete the SV.
- JBS p⁺ grid geometry and edge termination.

Critical: our V_{fd} and $W(0\text{ V})$ numbers assume p-n junction electrostatics. A Schottky/JBS front contact has a different built-in potential, so the bias required to fully deplete the same geometry will differ. The fab must re-derive the depletion bias from its measured barrier height and C-V. The SV geometry is portable; the bias numbers in §2 are not — treat them as the p-n reference, not a spec for the Schottky/JBS device.

5. Caveats — do NOT read as committed specs

- No Schottky/JBS electrical parameters — barrier height, leakage, C-V, turn-on are not modeled; do not expect them here.
 - Dark current / leakage: not specified. Our p-n leakage estimate is a single-point calibration and is irrelevant to a Schottky/JBS junction. The fab must measure leakage for its junction type.
 - No rated breakdown / max voltage. -50 V is solver convergence, not breakdown.
 - Tissue-equivalence (κ) and microdosimetric spectra are NOT in this spec. The tissue-equivalence correction depends on stopping-power data we have flagged as not foundry-ready (Phase 27 work). This document is geometry only; it makes no claim about lineal-energy spectra.
 - SV widths other than $50\text{ }\mu\text{m}$ half-width are not quoted — we only validated CCE uniformity at that width. Re-run uniformity sweeps before quoting other widths.
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6. What is solid (geometry hand-off-ready)

The SV depth, the full-depletion voltage and operating-bias window (for p-n electrostatics), the CCE-uniformity-optimized geometry, and $CCE \rightarrow 1$ when fully depleted. These are defensible TCAD geometry targets. The junction electrical behavior is the foundry's to define and measure.